

Project Milestones & Engagement Plan

This Engagement Plan outlines the planned sequence of activities, major milestones, estimated effort, and key outputs for the Hospital Readmission Reduction Analytics consulting engagement. The engagement progresses from business discovery and data assessment through analytical modeling, dashboard development, recommendations, and final executive delivery.

Major Project Milestones

Phase	Milestone	Major Activities	Key Output / Deliverable
1. Discovery	M1 — Project Initiation & Business Discovery	Define the readmission problem; identify business objectives; define key questions and KPIs; establish project scope and stakeholder perspective	Business Discovery Report
2. Data Assessment	M2 — Dataset Selection & Data Audit	Review healthcare dataset structure; assess completeness, missing values, data types, consistency, validity, and limitations; identify fields relevant to 30-day readmission	Data Quality Assessment
3. Data Engineering	M3 — SQL ETL & Data Preparation	Extract and transform data using SQL; clean and standardize fields; create derived variables; validate records; prepare analytical dataset for downstream analysis	SQL ETL Scripts + Clean Analytical Dataset
4. Analytics	M4 — Exploratory & Statistical Analysis	Examine readmission patterns; compare patient and clinical characteristics; analyze diagnosis categories and burden contributors; conduct statistical tests; identify meaningful relationships and patterns	Exploratory & Statistical Analysis

5. Predictive Analytics	M5 — Predictive Modeling & Risk Analysis	Develop and evaluate readmission prediction models; compare model performance; examine feature importance/relationships; assess limitations such as class imbalance and sensitivity	Predictive Modeling Report
6. Business Intelligence	M6 — Executive Dashboard Development	Develop Tableau KPIs and visualizations; analyze readmission trends and contributors; create interactive filters/actions; organize findings for executive consumption	Executive Tableau Dashboard
7. Insights & Recommendations	M7 — Executive Recommendations	Translate analytical findings into business/operational insights; identify key burden contributors; develop data-driven recommendations; distinguish actionable findings from clinical interpretation	Executive Recommendations Report
8. Engagement Closeout	M8 — Final Consulting Report & Presentation	Integrate methodology, findings, modeling results, dashboard insights, recommendations, limitations, and next steps into final executive communication	Final Consulting Report + Executive Presentation

Planned Engagement Timeline

Phase	Week 1	Week 2	Week 3	Week 4	Week 5
Discovery	■				
Data Assessment	■	■			
Data Engineering		■	■		
Analytics			■	■	
Predictive Modeling				■	
Dashboard Development				■	■
Recommendations					■

Final Reporting & Presentation					■
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Note: Timeline represents an illustrative consulting engagement sequence. Analytical activities may overlap and may be revisited as findings and data-quality issues emerge.

Estimated Effort

Phase	Primary Focus	Estimated Hours	% of Total
Business Discovery	Business objectives, stakeholder requirements, scope, KPIs	8	10%
Data Assessment & Engineering	Data audit, SQL ETL, cleaning, transformation, validation	16	20%
Exploratory & Statistical Analysis	EDA, readmission patterns, statistical testing, interpretation	20	25%
Predictive Modeling	Model development, evaluation, interpretation, limitations	16	20%
Dashboard Development	Tableau KPIs, visualization, interactivity, executive design	12	15%
Reporting & Presentation	Recommendations, final report, executive presentation	8	10%
TOTAL		80 hours	100%